Data Analytics Training

An Industrial Training Report Submitted To

Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal

Towards Partial Fulfilment for the Award of

Bachelor of Technology
In
Computer Science and Engineering

Submitted By

Saloni Shelukar(0820CS231102)

Department of Computer Science and Engineering
Lakshmi Narain College of Technology, Bhawarasla, Sanwer Road, Indore

Submitted To:

Training I/C
Mr.Arvind Mishra
(Techlene)

Mr. Pradeep Pal
HOD-CSE/IT
LNCT Indore

CERTIFICATE

*This is to certified that Training Report entitled "**Data AnalyticsTraining**" submitted by **Saloni Shelukar (0820CS231102)** is a satisfactory account of the bona fide work done in **'Techlene'** and is recommended for the Industrial Training, carried out in 5th Semester towards the partial fulfillment for the award of the degree **Bachelor of Technology** in **Computer Science and Engineering** to Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (MP.)*

Date: Endorsed
By:

Training I/C
Arvind Mishra
(*Techlene*)

Mr. Pradeep Pal
HOD-CSE/IT
LNCT Indore

TRAINING REPORT APPROVAL SHEET

*The training work entitled "**Data AnalyticsTraining**" submitted by **Saloni Shelukar (0820CS231102)**, is approved for Industrial Training as partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering by **Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P.).***

Internal Examiner **External Examiner**

STUDENT DECLARATION

I, ***Saloni Shelukar*** hereby declare that the presented report of Internship titled **"Data AnalyticsTraining"** at **Techlene** is uniquely prepared by me after the completion of three months of work/training at the **Techlene**.

I also confirm that this report is prepared solely for my academic requirements and not for any other purpose. It will not be used in the interest of any third party or corporation.

Saloni Shelukar (0820CS231102)

CSE (Department of Computer Science)
LNCT, Indore

Techlene Software Solutions Pvt. Ltd.
P101-102, Metro Tower,Near VijayNagar Square. INDORE
452 010 (M.P.) INDIA. Phone: +91 7047575000, Email :
info@techlene.com. www.techlene.com

Date : 30 October 2025

Dear **Saloni Shelukar**,

On behalf of **Techlene Software Solutions Pvt. Ltd.**, I am excited to extend an offer to you for an internship position within our development department. This position is located in Indore, Madhya Pradesh, and is specifically for the role of a **Data Analyst Intern**.

The internship is scheduled to begin on **November 3rd, 2025**, and will be a **month unpaid internship** ending on **December 17th, 2025**. The schedule for this position is **Monday to Friday**, from **10:00 AM to 5:00 PM**. During this period, you will report directly to **Mr. Arvind Mishra,** who will serve as your Reporting Manager.

As a temporary intern, you will not be eligible for any company-sponsored benefits. We ask that you bring two forms of ID.

During your temporary engagement with Techlene Software Solutions Pvt. Ltd., you may have access to confidential or proprietary business information belonging to the company. By accepting this offer, you acknowledge that this information must remain confidential and agree to refrain from using it for your purposes or disclosing it to anyone outside of Techlene Software Solutions Pvt. Ltd. Upon completion of your internship, you agree to return any company-issued property and documents promptly.

NOTE:

- **You will use your Laptop during this period.**
- **Any leaves you take during your internship will result in extension of your internship period accordingly.**

This offer letter represents the full extent of the internship offer and supersedes any prior discussions about the position. Any changes to this agreement must be made in writing. Please review this letter infull and sign and return it via email to the **HR Department** to confirm your acceptance.

We look forward to having you begin your career at Techlene Software Solutions Pvt. Ltd. and wish you a successful and enriching internship.

Welcome to the team!

For : Techlene Software Solutions Pvt. Ltd.

Dharni Singh
HR Manager

Acknowledgment

Every endeavor requires an indomitable urge, perseverance, and proper guidance, especially when it is most needed. Motivated to enhance my skills and knowledge, I embarked on this training program with hope and determination. The journey has been both enlightening and rewarding.

I express my heartfelt gratitude to my mentors, for their exceptional guidance, invaluable insights, and unwavering support throughout the training program. Their mentorship has been instrumental in shaping my understanding of Data AnalyticsDevelopment and applying it to practical scenarios.

Additionally, I would like to thank **Mr. Pradeep Pal (HOD, CSE/IT)** and **Dr. Prakash Bharani (Principal, LNCT Indore)** for their support and for fostering an environment conducive to learning. My gratitude also extends to the teaching and non-teaching staff of the Department of Computer Science and Engineering/Information Technology for their assistance throughout this training period.

I am deeply indebted to **Shri. Suprabhat Chauksey (M.D, LNCT Indore)** for providing the facilities and infrastructure necessary to successfully complete this training.

Lastly, I express my profound gratitude to my parents and my entire family for their constant encouragement and support. Without their dedication, this training would not have been possible. I also thank everyone who contributed, directly or indirectly, to the successful completion of this training.

It has been a privilege to learn and grow through this training program, which has added immense value to my academic and professional journey.

Saloni Shelukar

(0820CS231102)

Executive Summary

This project focuses on analyzing structured sales data to extract meaningful insights using **Data Analytics techniques**. The objective of the project is to identify sales trends, customer behavior patterns, and product performance through systematic data processing and visualization. The project demonstrates the complete data analytics lifecycle, including **data collection, data cleaning, exploratory data analysis, and result interpretation**. By applying statistical analysis and visual representation techniques, the system converts raw data into actionable information. This project provides practical exposure to data-driven decision-making and highlights the importance of analytics in business and organizational growth.

Table of Contents

1. **Training Report Approval Sheet**
2. **Certificate**
3. **Student Declaration**
4. **Certificate of Training (Internship)**
5. **Acknowledgement**
6. **Detailed Executive Summary**
7. **Comprehensive Introduction**
8. **Technical Background Study**
9. **Extended Requirement Analysis**
10. **Modular System Design**
11. **Technical Implementation Details**
12. **Rigorous Testing Protocol**
13. **Analysis of Results**
14. **Conclusion & Future Roadmap**

Introduction

Data Analytics plays a vital role in modern organizations by enabling informed decision-making based on historical data. Businesses generate large volumes of data daily, but without proper analysis, this data holds little value.
This project aims to analyze sales data to uncover hidden patterns, trends, and insights that can support strategic planning. The focus is on transforming raw data into meaningful information using analytical techniques and visual tools.

Background Study

Traditionally, business decisions were made based on experience and intuition. However, with the rapid growth of digital systems, data-driven approaches have become essential.

This project explores the fundamentals of **data analytics**, including data preprocessing, statistical analysis, and visualization. It also studies how analytical insights help organizations optimize sales performance, improve customer satisfaction, and increase revenue.

Requirement Analysis

Hardware Requirements

● Processor: Any modern system capable of handling data processing
● Memory (RAM): Minimum 4 GB
● Disk Space: 200 MB or more

Software Requirements

● Operating System: Windows / Linux / macOS
● Tools: Python / Excel / Power BI / Tableau (any one)
● Dataset: Structured sales dataset (CSV format)

Functional Requirements

● Import and preprocess sales data
● Handle missing or inconsistent data
● Perform statistical analysis
● Generate visual reports and charts
● Interpret and summarize analytical results

System Design

The project follows a **data analytics workflow model**:

1. **Data Collection**
   - Acquire data from structured sources
2. **Data Cleaning & Preparation**
   - Remove duplicates and handle missing values
3. **Exploratory Data Analysis (EDA)**
   - Analyze trends, patterns, and relationships
4. **Visualization & Reporting**
   - Present insights using graphs and dashboards

Implementation

The implementation phase includes:

● Data loading and preprocessing
● Descriptive statistical analysis

- Trend and pattern identification
- Visualization using charts such as bar graphs, line charts, and pie charts
- Summary generation based on insights

Testing

The project was tested to ensure accuracy and reliability:

- Data integrity testing
- Validation of analytical results
- Visualization correctness testing
- Cross-checking insights with raw data

Result

The project successfully identified sales trends, top-performing products, and seasonal variations. The visual reports made complex data easy to understand and supported data-driven decision-making.

Conclusion

The **Sales Data Analysis Project** demonstrates the practical application of data analytics in solving real-world problems. It enhanced understanding of data processing, analysis techniques, and visualization methods.

Future Roadmap

- Predictive analytics using machine learning
- Real-time data analysis
- Integration with larger datasets
- Automated dashboard generation